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Textile –Global

✓The textile industry or apparel industry is primarily concerned with the design and production of yarn, cloth, clothing, and their distribution. The raw material may be natural, or synthetic using products of the chemical industry.

✓The current global garment market is approximately US$ 1.15 trillion which forms nearly 1.8% of the world GDP. Almost 75% of this market is concentrated in EU-27, USA, China and Japan.

✓China is the largest textile producing and exporting country in the world.

✓In 2014, China's textile exports were valued at just over 110 billion U.S. Dollars.

✓The world garment market is growing at a CAGR of 5% and attaining a size of about US$ 2.21 trillion by 2025

✓Textile and Clothing nominal sales ceded -4% in 2015 and were crippled by unusually low commodity prices (cotton -15%, wool -7%, and manmade fibers 1%) and multiple currency depreciations worldwide, particularly in emerging markets.

Source: : World Statistics , Textile Ministry, Make in India, TechSci Researh
Note: Figures are as per latest data available

Countries	Market Size in n. USD 2013	Market Size in n. USD 2025	CAGR %
Europe	355	440	2
USA	230	285	2
China	165	540	10
Japan	110	150	2
Brazil	60	100	5
India	46	200	12
Russia	45	105	8
Canada	30	50	4
Australia	25	45	5
RoW	80	195	8
Total	1146	2210	5

Textile – India on Global Stage

✓According to UN Comtrade data released in June, 2014

India was announced as the world's second largest exporter of textiles and clothing in the world

And 6th largest exporter of clothing only

✓India is the 2nd biggest producer of silk and cotton

✓Indian textile industry accounts for about 24 per cent of the world's spindle capacity and 8 per cent of global rotor capacity

✓India has the highest loom capacity (including hand looms) with 63 per cent of the world's market share

✓India accounts for about 14 per cent of the world's production of textile fibres and yarns (largest producer of jute, second largest producer of silk and cotton; and third largest in cellulosic fibre)

Source: Textile Ministry, Make in India, TechSci Research

Note: Figures are as per latest data available

Key Facts- Textile in India

The fundamental strength of the textile industry in India is its strong production base of wide range of fibre/yarns from natural fibres like cotton, jute, silk and wool to synthetic /man-made fibres like polyester, viscose, nylon and acrylic.

✓Textile contributes to 3-4% to country's gross domestic product.

✓17% to country's export earnings come from Textile.

✓Provides direct employment to over 35 million people in India.

✓It is the largest provider of employment after agriculture.

✓India earns about 27% of its total foreign exchange through textile exports.

✓The textile industry of India also contributes nearly 14% of the total industrial production of the country.

✓Among the countries, Japan, Australia and New Zealand are the significant consumers of Indian textiles.

Source: Textile Ministry, Make in India, TechSci Research
Note: Figures are as per latest data available

The Sector Has Been Posting Strong Growth Over The Years

- Textile plays a major role in the Indian economy
- It contributes 14 per cent to industrial production and 4 per cent to GDP
- With over 45 million people, the industry is one of the largest source of employment generation in the country
- The size of India's textile market in 2014 was USD99.0 billion which is expected to touch USD226 billion market by 2023 at a CAGR of 8.7 per cent between 2009-23E



Source: Textile Ministry, Make in India, TechSci Research
Note: Figures are as per latest data available

TEXTILE INDUSTRY

Organised/ Centralized mills sector

a. Spinning Mills
b. Composite mills
(Spinning and weaving)

Unorganised/ Decentralized mills sector

a. Hand Looms
b. Power Looms
c. Hosiery and Khaddi units
d. Processing units

SEGMENTS

Indian textile industry can be divided into several segments, some of which can be listed as below:

- Cotton Textiles
- Silk Textiles
- Woollen Textiles
- Readymade Garments
- Hand-crafted Textiles
- Jute and Coir

WELSPUN INDIA: WORLD'S LARGEST HOME TEXTILE COMPANY

✓Welspun India Ltd. (WIL) is a fully integrated home textile manufacturer.

✓It is one of the largest global home textile players, with world class manufacturing facilities in India.

✓Ranked 1st in the Top 15 Supplier Giants (USA) by Home Textiles Today Magazine, January 2013.

✓Anjar facility is the largest home textiles facility in Asia.

✓International operations has been the major part of the revenues.

✓ Innovative products accounted for 31% of the revenues

✓Increased vertical integration for yarn and fabric to ~70%

WELSPUN BRANDS

SPACES
HOME & BEYOND

Premium brand with presence in India and Middle East.

Footprint of over 200 shop-in-shop outlets and distribution

CHRISTY
Est.1850

Largest terry towel brand in the UK with growing presence in China and the Middle East

Present in over 40 retailers within UK

Also present in USA and seeing aggressive growth in e-commerce

WELSPUN
WELHOME
BED | BATH | KIDS | LIVING

Mass brand presence via mass retailers; also growing via distributor network

KINGSLEY
HOME

Mass brand in the UK with a focus on bedding Growing aggressively in affordable fashion

Biggest Client Across Globe



Wall Mart
Bed Bath & Beyond
Macy's

J C Penney
Target
K Mart- Sears



ASDA
Tesco
Marks & Spencer

Debenhams
House of Frazer
Christy



Ikea
Carrefour
JYSK

El Corte Ingles
Vincenzo Zucchi S.p.A



Spaces- Home & Beyond
Welhome

GLOBAL PRESENCE



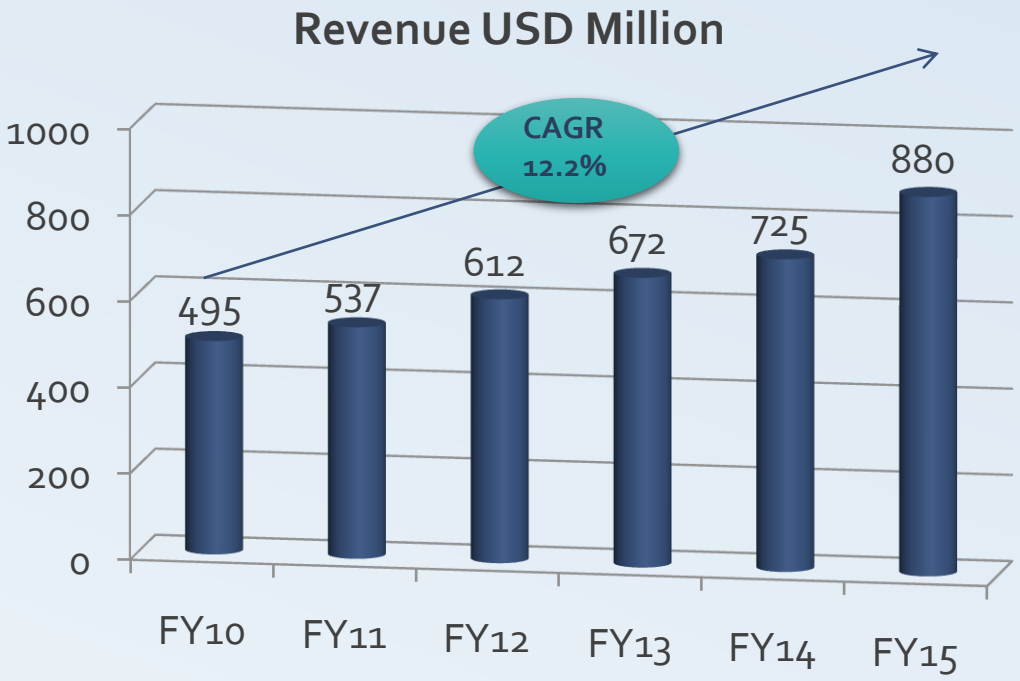
WELSPUN INDIA: WORLD'S LARGEST HOME TEXTILE COMPANY

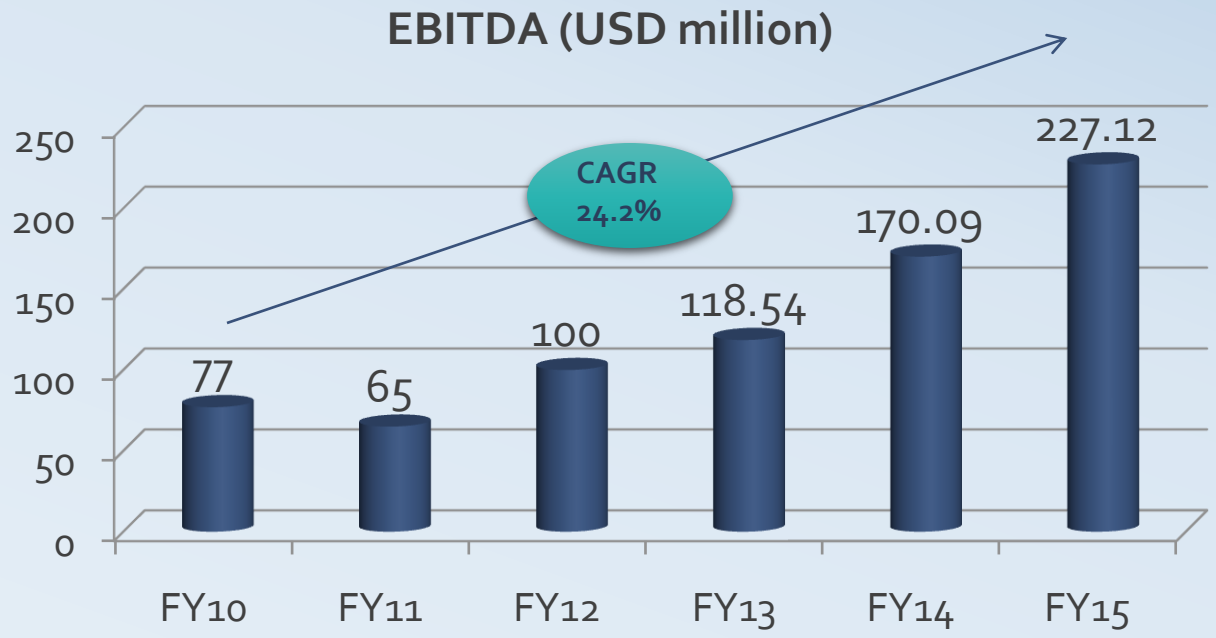


Terry towels	Bed linen products	Rugs
Capacity – 50,000 MT/Year	Capacity – 60 million metre/Year	Capacity – 15,000 MT/Year
Location - Anjar/Vapi	Location - Anjar	Location - Vapi
Capacity utilization - 102 per cent	Capacity utilization - 97 per cent	Capacity utilization - 58 per cent

WELSPUN INDIA: WORLD'S LARGEST HOME TEXTILE COMPANY

- ✓Welspun ranked 1st in home textile supplies to US
- ✓Company exported 97 per cent of the production in FY15
- ✓During FY10-15, revenue of Welspun increased at a CAGR of 12.2 per cent in USD terms
- ✓During FY10-15, EBITDA of Welspun increased at a CAGR of 24.2 per cent in USD terms





Company	Business areas
Welspun India Ltd	Home textiles, bathrobes, terry towels
Vardhman Group	Yarn, fabric, sewing threads, acrylic fiber
Alok Industries Ltd	Home textiles, woven and knitted apparel fabric, garments and polyester yarn
Raymond Ltd	Worsted suiting, tailored clothing, denim, shirting, woollen outerwear
Arvind Mills Ltd	Spinning, weaving, processing and garment production (denims, shirting, khakis and knitwear)
Bombay Dyeing & Manufacturing Company Ltd	Bed linen, towels, furnishings, fabric for suits, shirts, dresses and saris in cotton and polyester blends
Garden Silk Mills Ltd	Dyed and printed fabric

Competition

(Rs. cr.)

Company	Total Assets	Net Profit	Employees and Operations	Corporate Office
Alok Industries	~18,000	601.74	20,000+ and 250 Stores in India, it also exports to more than 20 countries	*Mumbai, Maharashtra*
Wardhman Textiles	5507	561	23,000 to 25,000 and exports in 35 different countries	Ludhiana, Punjab
Arvind Mills	*2500 - 3000*		25,000 to 28,000 with over 1000 counters for selling	Ahmadabad, Gujarat
Welspun India	3,449	601	25,000+ and Exports to 34 countries and supplies to 12 out 20 retail chains	Mumbai, Maharashtra
Bombay Dyeing	1674		10,000 to 12,000 employees and 350+ Retail Chains, 2000+ Multi Brand Stores	Mumbai, Maharashtra
Raymonds	~1500		10,000 to 12,000 with over 700 retail shops spread over 200 cities in India and aboard	

Competition

(Rs. cr.)

Name	Market Cap. Mar'15	Sales Turnover Mar'15	Net Profit Mar'15	Total Assets Mar'15
Welspun India	11,011.79	4,867.94	601.74	3,449.26
Siyaram Silk	1,026.36	1,619.36	87.57	715.42
Alok Industries	606.02	11,752.39	-3,722.80	18,683.41
Orbit Exports	358.78	148.46	24.53	135.78
E-Land Apparel	179.96	234.55	-18.67	166.47
Garden Silk	154.86	2,375.48	-140.61	1,219.33
Donear Ind	153.66	517.29	5.63	363.87
BSL	85.48	378.27	7.12	192.24
Arrow Textiles	83.03	49.13	5.31	40.61
Pradip Overseas	6.54	237.68	-280.98	1,137.11

Source: http://www.moneycontrol.com/competition/welspunindia/comparison/WI03

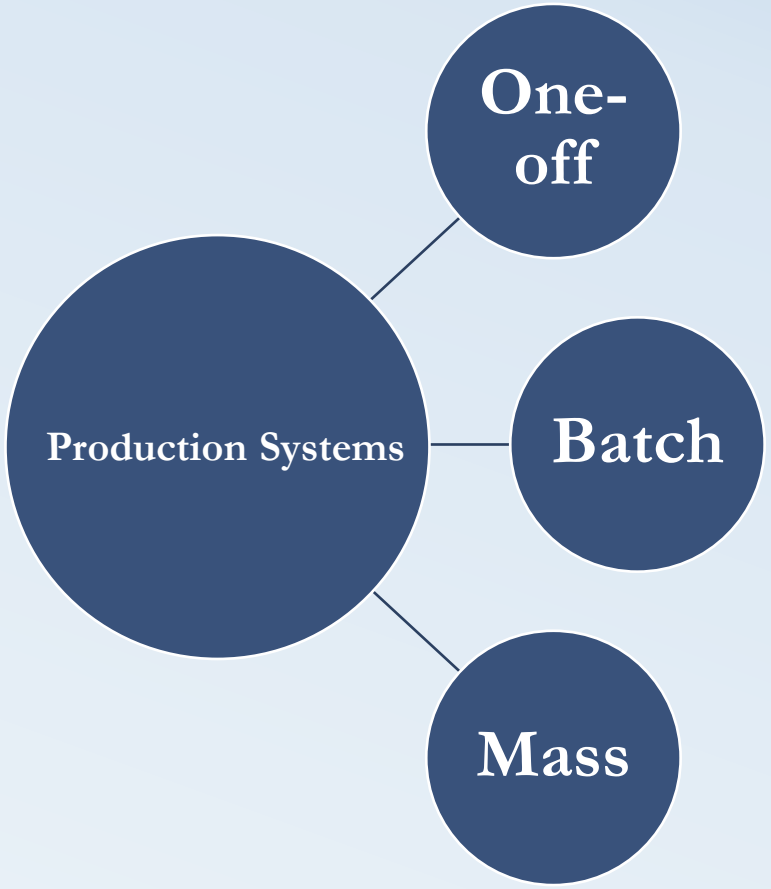
Textile Manufacturing process

- ✓Garment manufacturing is an assembly- oriented activity .
  - ➢Great range of raw materials
  - ➢product types
  - ➢production volumes
  - ➢supply chains
  - ➢retail markets
  - ➢associated technologies.

- ✓Companies range from small family business to multinationals.

- ✓The clothing industry is labour intensive industry.





One-off

- **One-off production** is designing and making a single textile product to a client's specification.
- The garment design is developed from a basic block pattern, with a Prototype made from inexpensive fabric to test the drape, fit and assembly of the garment.

Batch

- **Batch production** is manufacturing set quantities of a textile product to order.
- The quantity of products can vary from a set of four cushions made by a designer-maker, to 20,000 jumpers made for a department store

Mass

- **Mass production** is industrial-scale manufacture of large quantities of products, usually on a production line.
- Mass production is suitable for products that seldom need to be redesigned and are needed in very large numbers, eg socks or jeans.

Production system	Product market	Design and production	Skill level and cost
One-off	Made-to-measure. eg suit, wedding dress;	Made-to-measure garments are made to fit the measurements of an individual client; the garment design is developed from a basic block patterns and a toile is made to test the fabric drape, the fit and order of assembly	Very high-level skills in design and manufacture; high-cost materials; high labour costs
One-off	Haute Couture. eg made by fashion houses	Fashion designers such as John Galliano design Haute Couture garments for individual clients	Very high-level skills in design and manufacture; high-cost material and labour costs
Batch production	Ready-to-wear (RTW) designer label eg Designers at Debenhams	Garments are designed to fit a range of standard sizes and shapes. Garment patterns are developed from a basic block using CAD. A sample garment is made up in a medium size, from the intended fabric. Once the design has been approved it is put into production in a range of standard sizes. They are sold through up-market retailers.	High-level design, pattern making and sampling skills; cost-effective materials and lower manufacturing costs
Mass production	Mass-market retailers,. eg Top Shop	Similar production methods to batch production: garments produced in limited range of sizes; standardised production methods are used to produce a wide range of styles. Most fashion products are batch produced in large batches eg 20,000. Some classic products like jeans are mass produced for a world market.	High-level design, pattern making and sampling skills; cost-effective materials

CAD

- ✓ USING COMPUTERS TO DESIGN AND PRESENT
- ✓ Computers are used by designers for:
- ✓ Writing documents and creating display boards, including artwork, text, spreadsheets, graphs and tables
- ✓ Putting together slide show presentations
- ✓ Digital photography and video making
- ✓ Designing and sampling
- ✓ Supplementing drawing and colouring with use of specialist fashion software, e.g. Speed Step.

Programmes used by designers:

PowerPoint, Publisher ,Word, Excel, paint

CorelDraw, Adobe illustrator, Photoshop

Specialist Fashion software- Speed Step

CAM

✓Computer-controlled machines /CAM play a vital role in modern textiles production. For example, they enable :

I. **designs to be sent electronically to the print manufacturer** and stored on computer to ease repeat printing orders.

II. **colours to be matched** to the design, dyes weighed and dispensed and the fabric printed automatically.

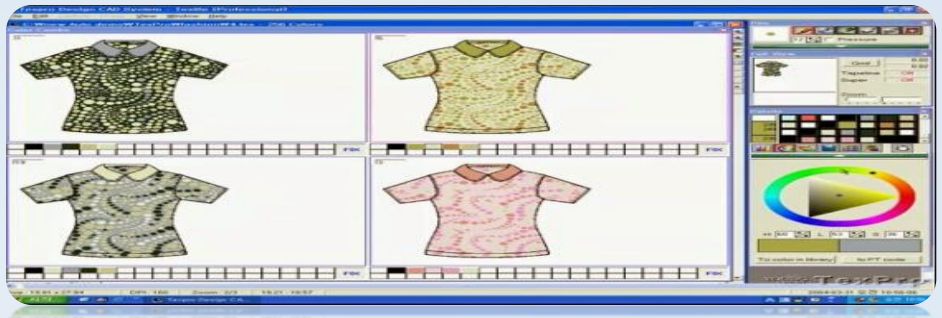
III. **print directly onto fabric**, or transfer a design from paper to fabric using heat. This can be extremely useful when manufacturing includes hazardous and toxic processes, such as dyeing.

IV. **load fabric onto the cutting machine** and operate the cutting equipment.

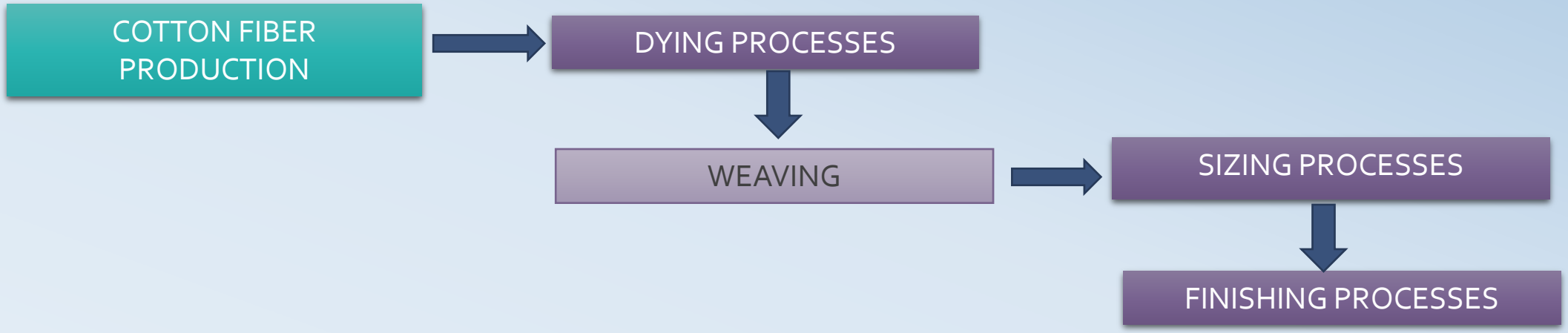
V. **stitch the pieces of fabric together** – automatic sewing machines include microprocessors to control the movement of pattern pieces, and sensors which stop the machine if anything goes wrong. CAM enables batch production to run smoothly as it does not take long to reset equipment as data is stored and saved.







Production Flow



Dyeing Processes



Source: http://www.textiletoday.com.bd/magazine/609

Cotton manufacturing

Cotton is the world's most important natural fibre. cultivated in more than 50 countries.

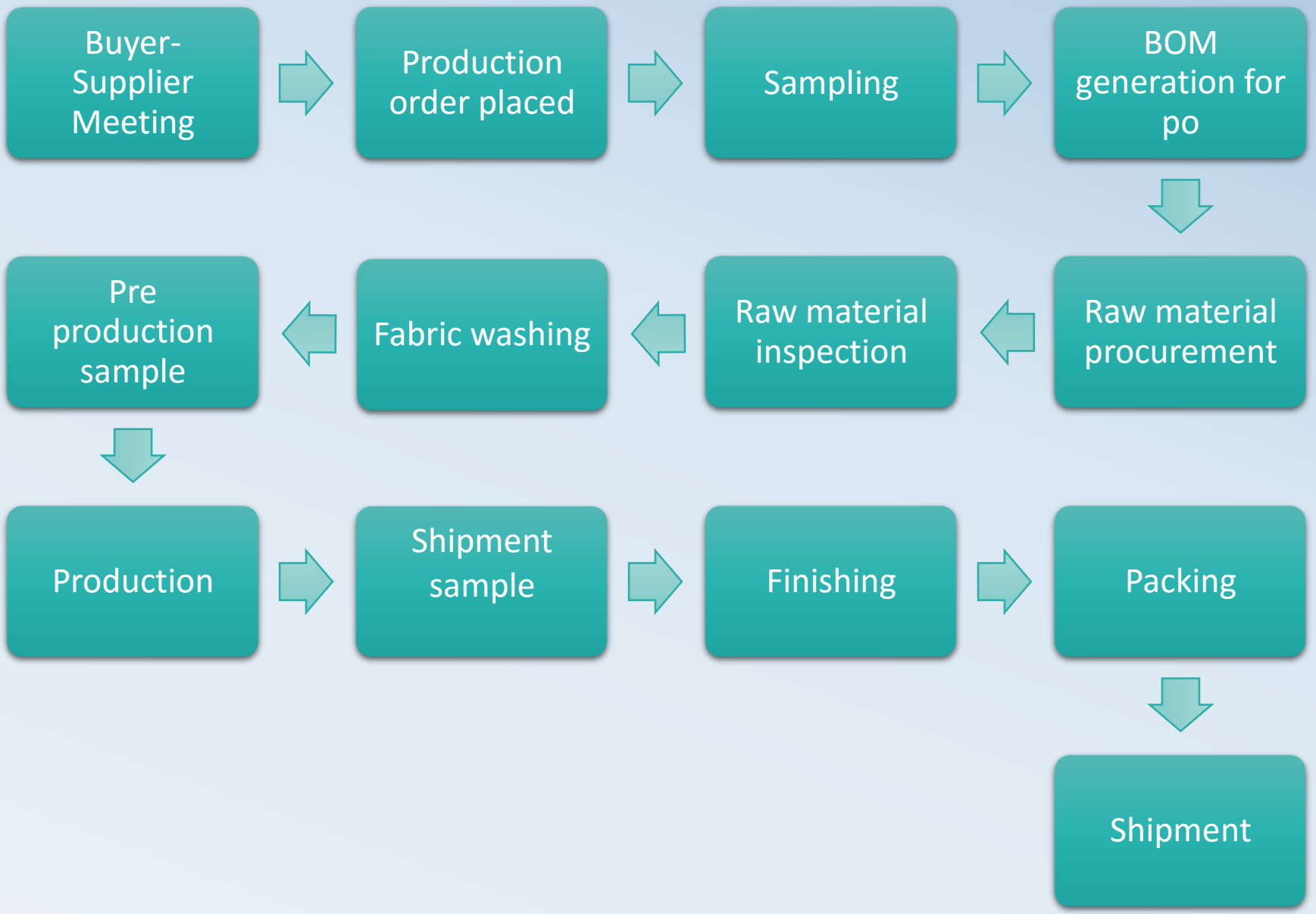
- ✓Cultivating and Harvesting
- ✓Preparatory Processes
- ✓Spinning- giving yarn
- ✓Weaving- giving fabrics
- ✓Finishing- giving textiles

Departments in garment manufacturing unit



Source: http://www.textiletoday.com.bd/magazine/609

Process flow in garment manufacturing unit



Source: http://www.textiletoday.com.bd/magazine/609

Merchandising department

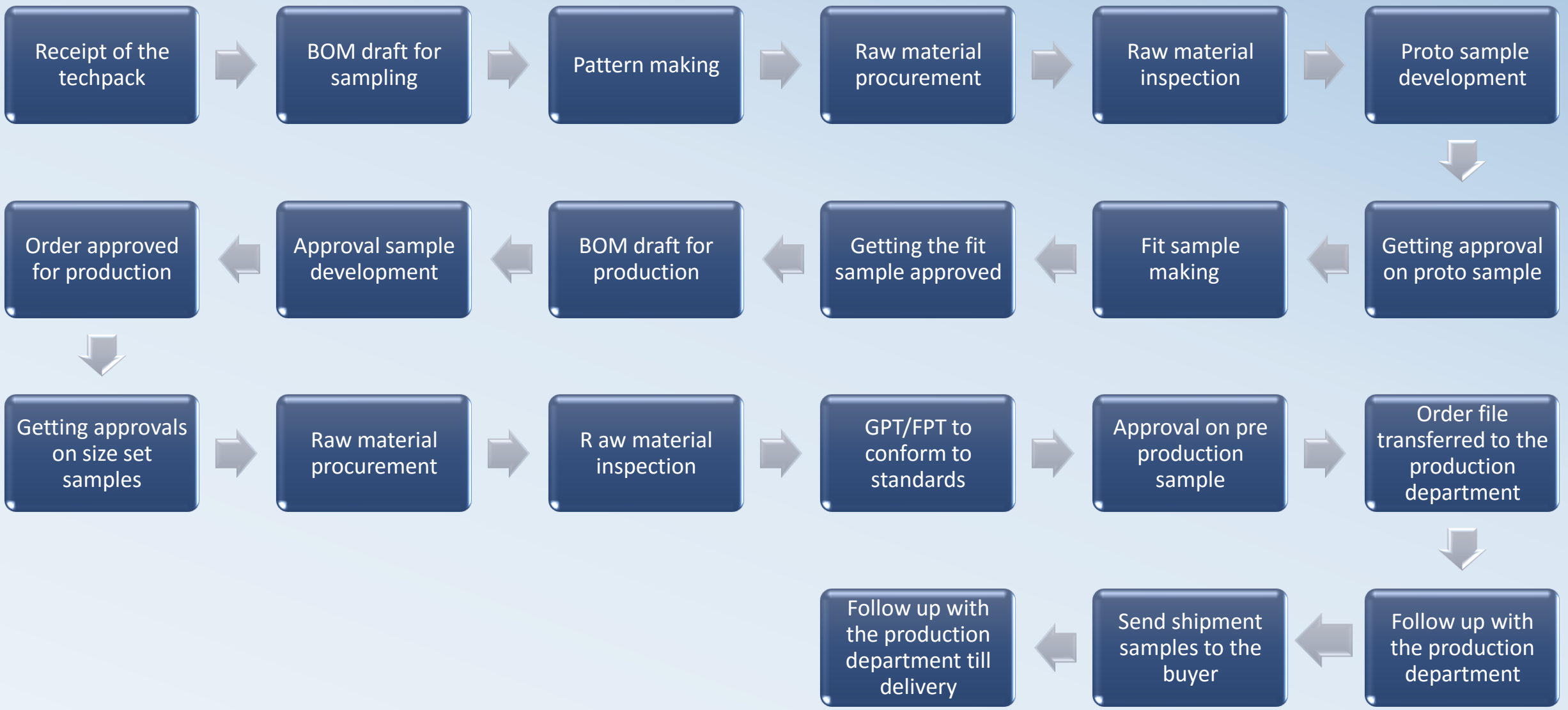








Process flow in merchandising department



Responsibilities of a merchandiser



Source: http://www.fibre2fashion.com/industry-article/36/3516/merchandising-in-an-apparel-industry1.asp

Sampling department







Responsibilities of sampling department

- Getting clarifications about style details from merchandiser.
- Checking pattern’s workability.
- Preparation of different samples and getting the buyer’s approval.
- Informing quality related problems, encountered during preparing samples, to QC.
- Minimizing operations and consumption.

Fabric store department









Functions of the fabric store department

- Receive the fabric as per BOM
- Fabric inspection (four point system)
- Conduct fabric test (e.g. Shrinkage, colour fastness etc.)
- Issue to the production department as per the requirement



Trims and accessories
Store department









Functions of the trims and accessories department

- To receive incoming material.
- To check material for attribute & variables.
- To arrange the incoming material in allocated racks.
- To issue trims as per BOM for sampling and production & other material as per requirement.
- To maintain a record of incoming & outgoing material.



Spreading and cutting department

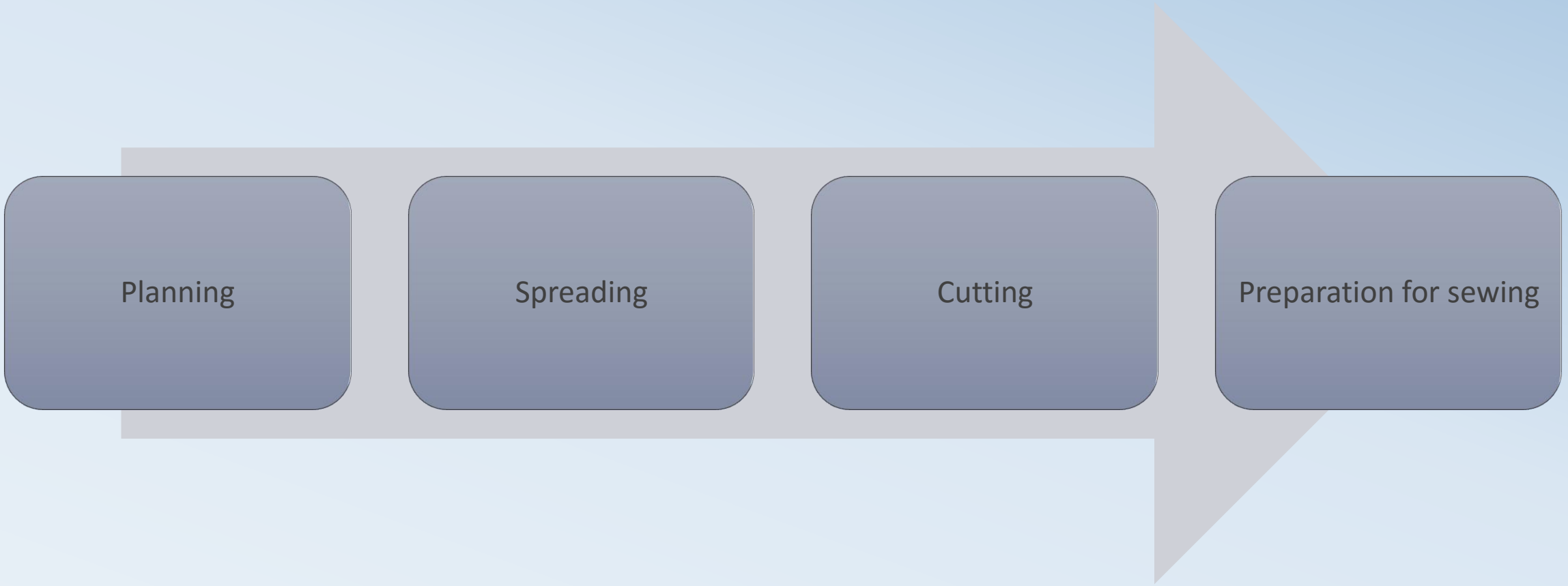




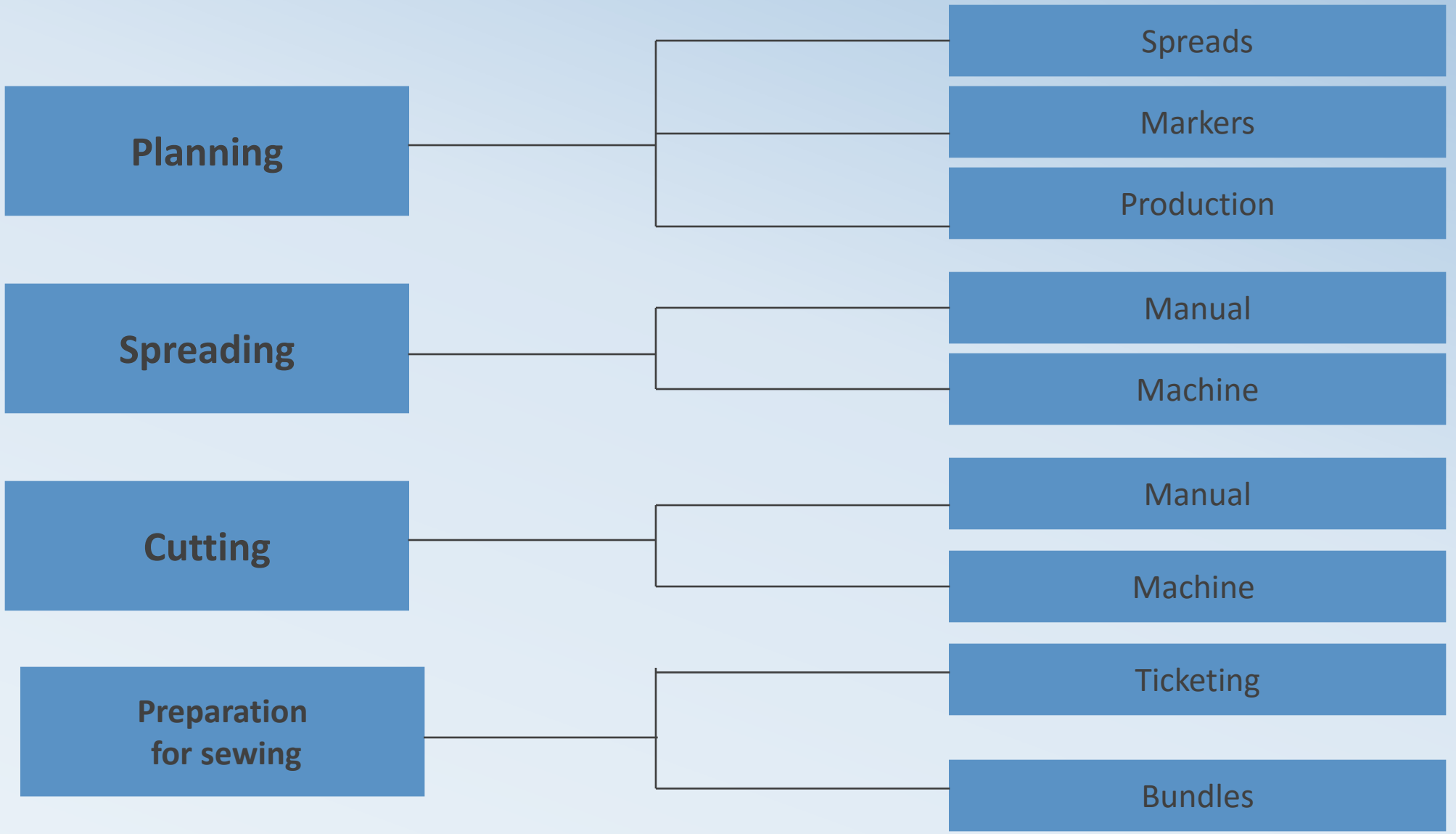




Spreading and cutting flow process



Spreading and cutting flow process

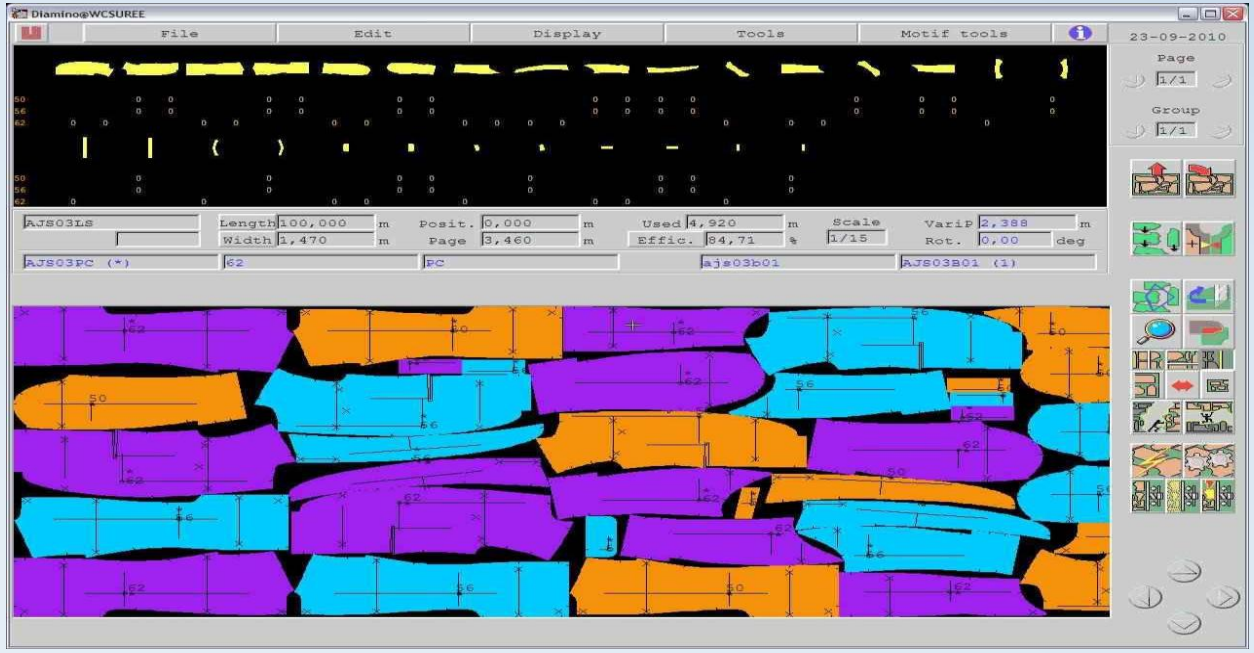


Marker

- is a diagram of a precise arrangement of pattern pieces for a specific style and the sizes to be cut from a single spread.

Marker making

- is the process of determining the most efficient layout of pattern pieces for a specified style, fabric, and distribution of sizes (requires time, skill and concentration)



Spreading

- Spreading is the processes of superimposing lengths of fabric on a spreading table cutting table or specially designed surface in preparation for the cutting process

- Spread or lay is the total amount of fabric prepared for a single marker

Requirement of spreading

- Shade sorting of cloth pieces
- Correct ply direction and adequate lay stability
- Alignment of plies
- Correct ply tension
- Elimination of fabric faults
- Avoidance of distortion in the spread

Spreading equipment

- Spreading surfaces (table, pin table, vacuum table)
- Spreading machines

Spreading

Spreading equipment

- Spreading surfaces (table, pin table, vacuum table)
- Spreading machines

Cutting room

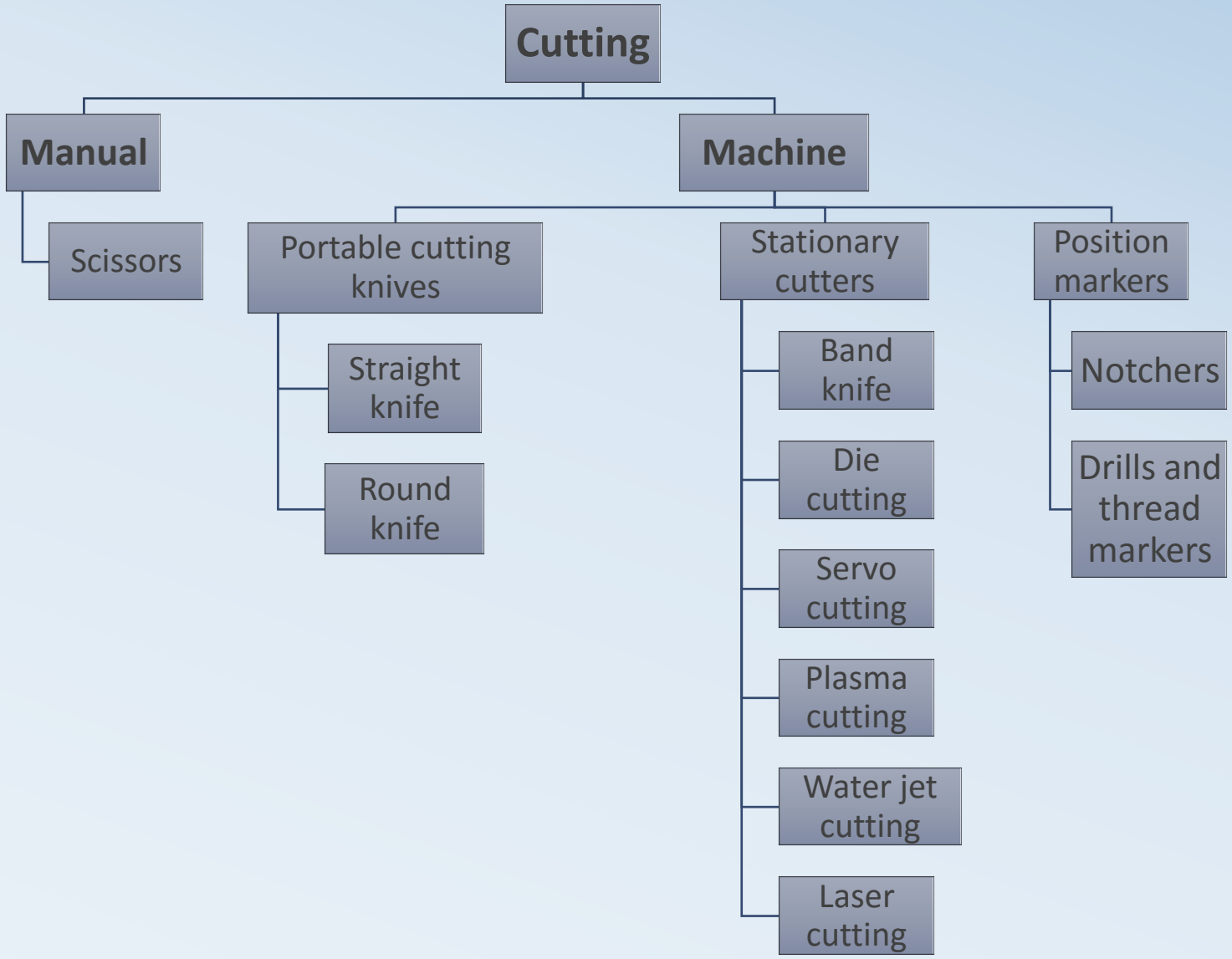






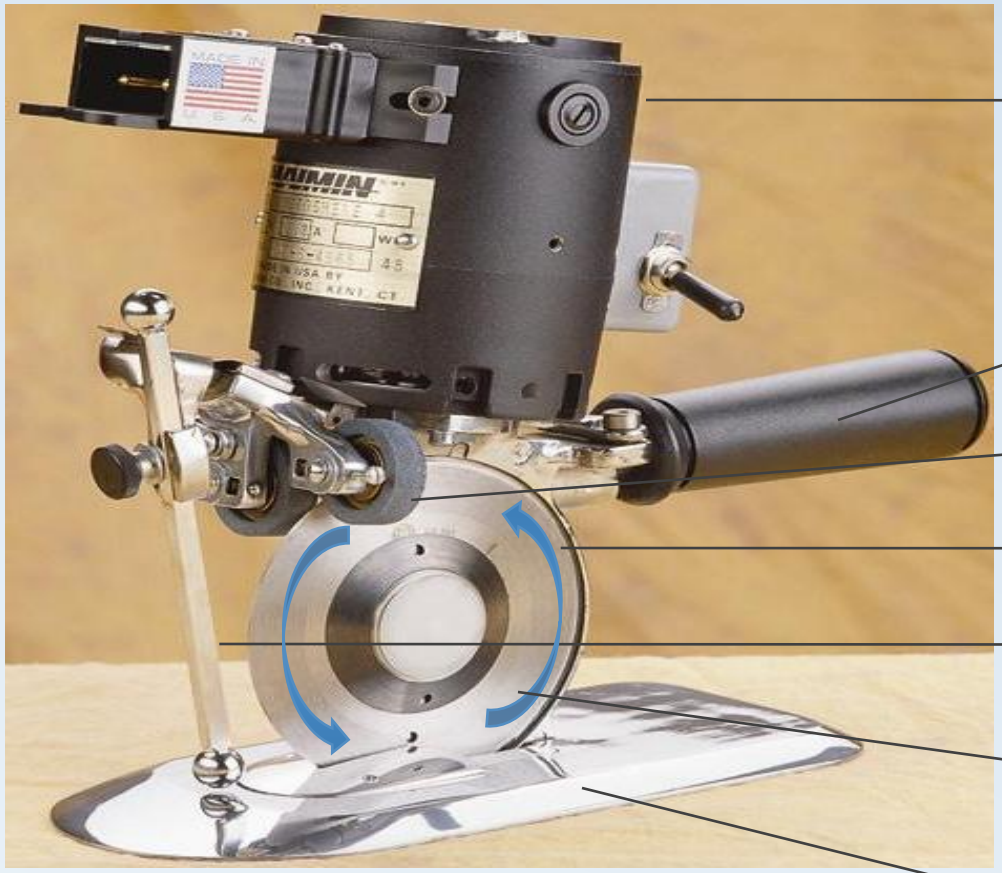


Cutting


Cutting
Manual
Scissors
Machine
Portable cutting knives
Straight knife
Round knife
Stationary cutters
Band knife
Die cutting
Servo cutting
Plasma cutting
Water jet cutting
Laser cutting
Position markers
Notchers
Drills and thread markers

Cutting

Portable cutting knives



Round knife

Power system

Handle

Sharpening

Cutting blade

Blade guard

Up and down movement

One way thrust as the circular blade makes contact with the fabric

Base plate



Straight knife

Sewing department

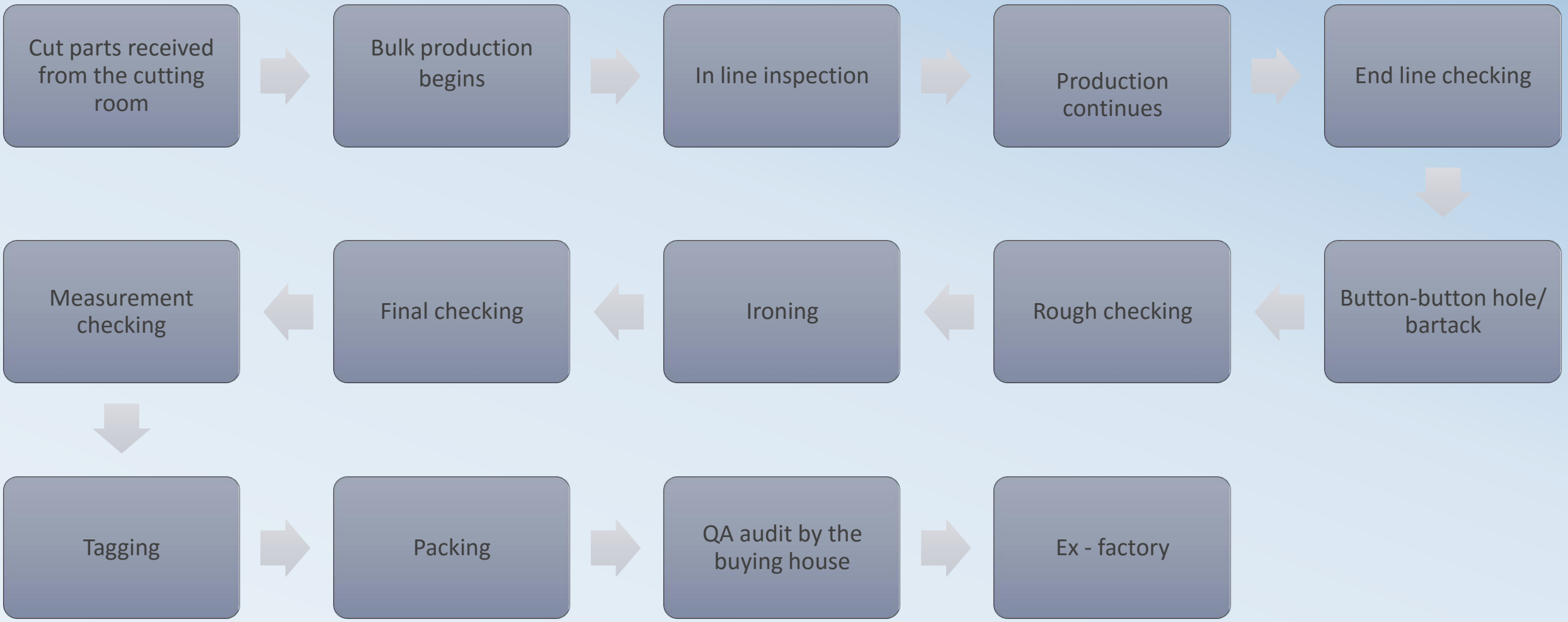








Sewing process flow



Sewing Machine

Machine beds

Raised bed



Flat bed



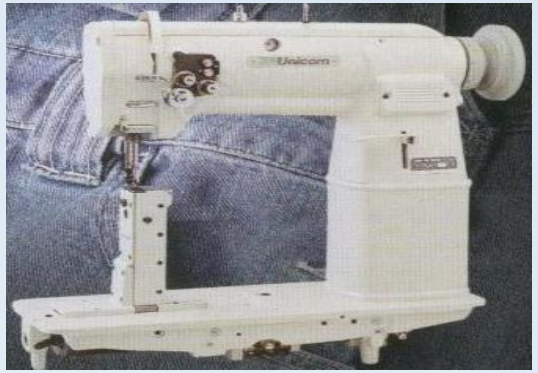
Cylinder bed



Feed of arm



Post bed



Source: Glock & Kunz (Third edition)

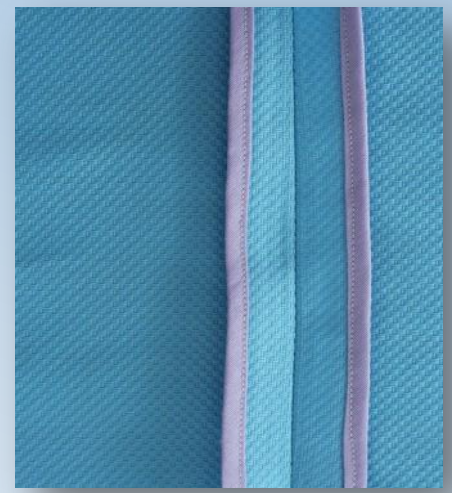
Seams & Stitches

Classification of seams

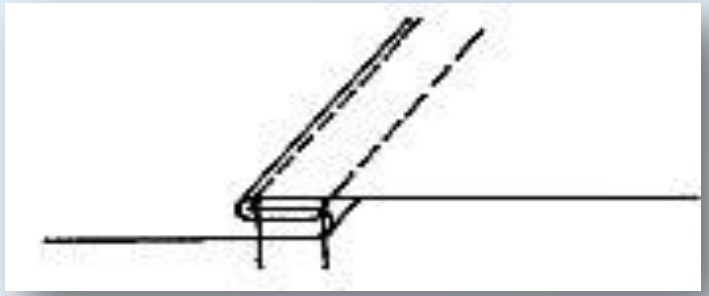
- Superimposed seam
- Lap seam
- Bound seam
- Flat seam
- Edge finishing



Superimposed seam



Bound seam



Lap seam



Edge finishing



Flat seam

Source: Glock & Kunz (Third edition)

Seams & Stitches

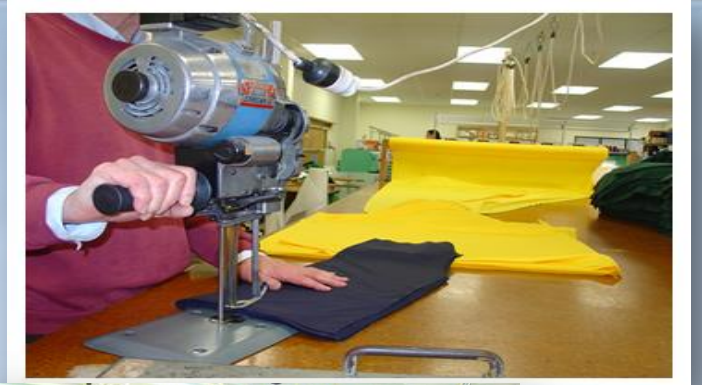
Stitch classification

- Stitch classification is based on structure of the stitch and method of interlacing.

100 Class stitch: Single thread chain stitch	• Using one needle thread and one blind looper
200 Class stitch: Single thread hand sewn stitch	• Using one needle thread
300 Class stitch: Two or more thread lock stitch	• Using Needle Thread(s) and One Bobbin Hook Thread
400 Class stitch: Multi-thread chain stitch	• Using one or more needle threads and one or more looper threads
500 Class Stitch: Multi-thread over edge chain stitch	• Using needle thread(s) and looper thread(s)
600 Class Stitch: Multi-thread cover stitches	

Source: Glock & Kunz (Third edition)

Quality department









Quality department

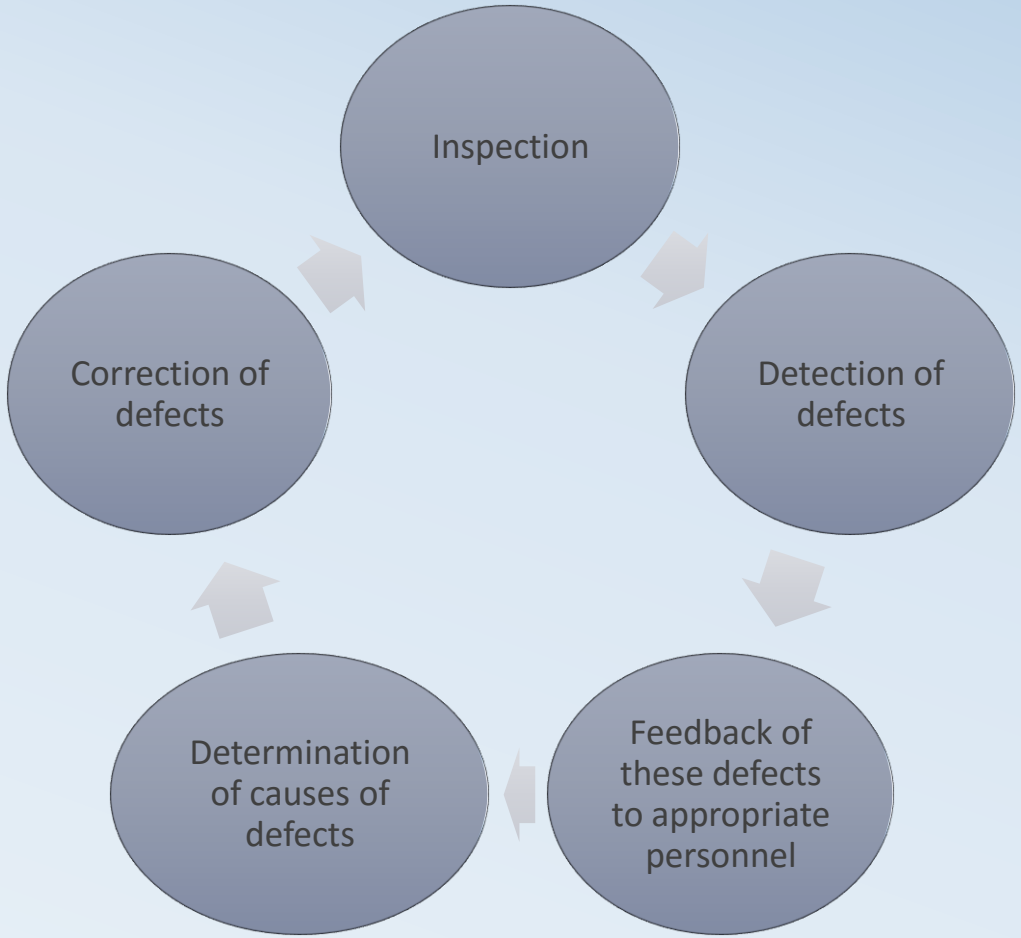
Responsibilities of quality department

- To impart quality in the product.
- To ensure that the product has achieved the quality parameters of buyers.
- To restrict the defects entering into the final product.
- Main function of quality department is to carry out *inspection.*
- *Inspection* can be defined as the visual examination or review of raw materials, partially finished components of the garments and completely finished garments in relation to some standards, specifications, or requirements, as well as measuring the garments to check if they meet the required measurements.

Source: P. B. Mehta and S. K. Bhardwaj, " Managing quality in apparel products"

Quality department

Principle of inspection (inspection loop)



Source: P. B. Mehta and S. K. Bhardwaj, " Managing quality in apparel products"

Quality department

Checks for final inspection

- Open seams
- Skipped stitches
- Cracked stitches
- Stitches/inch
- Uneven seams
- Crooked, puckered, curled, pleated seams
- Needle and feed cuts
- Unclipped threads and Long ends
- Raw edge
- Snaps, fasteners, buttons
- Labels
- Elastic
- Measurements
- Mends or repairs
- Stripe
- Hems
- Trim
- Broken needle
- Distortion

Source: P. B. Mehta and S. K. Bhardwaj, " Managing quality in apparel products"

Fabric audit procedure



- ~10% random inspection for sampling or lining or net fabric.
- Inspection for production order fabric as per buyer's requirement.
- Perform GSM cutting.
- Perform shrinkage test.
- Check the fabric for the bowing or skewing.
- Check fabric roll for the defects.
- Allocate points to the defects.



The fabrics are checked for:

- Width
- Shade (centre to selvedge)
- Weaving defects
- Shrinkage (steam & fusing)

Finishing and packaging department









Finishing and packaging department

- Finishing is the last stage of garment production where garment gets its final look.
- In this department each garment undergoes different finishing processes.
- It undergoes for quality check for several number of time which sets the garment free from defects.
- Buyer specifications and instructions are strictly maintained.

Source: P. B. Mehta and S. K. Bhardwaj, " Managing quality in apparel products"

Finishing and packaging department

Operations performed at finishing stage

- *Thread cutting:* Uncut threads affect the presentation of finished and packed garments. Therefore, it is necessary to cut and trim the loose and uncut threads.

- *Stain removal:* Removal of the following type of stains:

  Oil, yellow, black and paint spots, stains due to color bleeding, ink, rust, tracing marks, yellow stains, and hard stains

- Seam ironing: Ironing of garments using steam ironing tables with vacuum boards.

- *Final finishing: T*he entire garment is finished using various finishing equipments.

Source: P. B. Mehta and S. K. Bhardwaj, " Managing quality in apparel products"

Positions in Textile industry

Corporate positions

- ✓ Corporate Sourcing head
- ✓ HR Manager
- ✓ Manager – Accounts and Finance
- ✓ IT Operations Manager
- ✓ Divisional Merchandising Manager
- ✓ Senior Manager compliance.- Textiles
- ✓ Head Visual Merchandising
- ✓ Head Taxation
- ✓ Fabric sourcing manager

Plant and Store Positions

- ✓ General Manager Quality
- ✓ Senior manager engineering
- ✓ Senior Production Manager
- ✓ Warehouse- Incharge
- ✓ Manger - Velvet Weaving
- ✓ Senior Merchandiser
- ✓ Weaving & PPC Head
- ✓ Spinning Manager
- ✓ Technical Specialist - Dyes And chemicals
- ✓ Head of Dye House-Knit Fabrics

Sales and Marketing Positions

- ✓ Marketing Head(Apparel & Textile)
- ✓ Senior Regional Sales Manager
- ✓ Brand Manager
- ✓ VP Sales Exports
- ✓ Manager-Institutional Sales

R&D Positions

- ✓ Head Designer
- ✓ Manager- Design & Product
- ✓ Creative Head
- ✓ General Manager Product development

Thank you







